

Edge AI Computing for Grape Leaf Disease Detection

Thesis Presentation



UNIVERSITÀ DI PISA

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Advisories:

Prof. Vallati Carlo

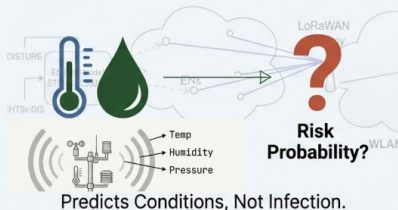
Dr. Palla Fabrizio

Prof. Zinnai Angela

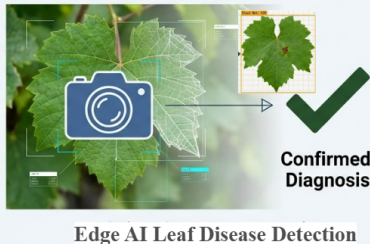
Candidate: *Gebrelibanos Tsegay*

Motivation → Beyond Environmental Data

Environmental Data Monitoring

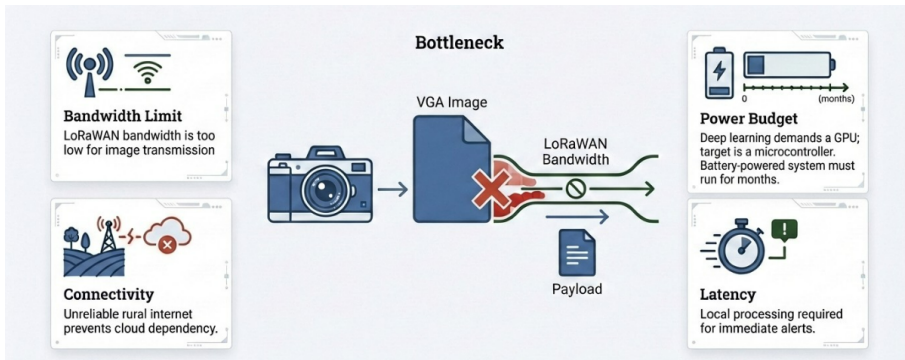


Thesis Goal (Direct Visual AI)



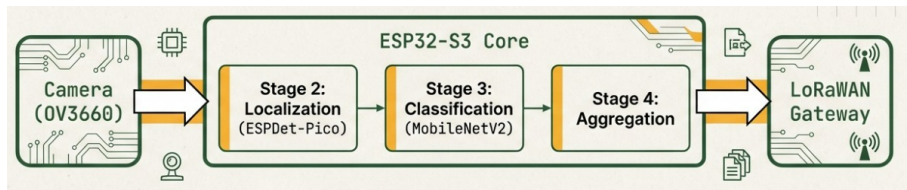
- *LoRaWAN-based WSNs are widely used for vineyard monitoring*
- *Environmental data predicts disease risk but doesn't confirm presence*
- *Visual-based edge detection confirms disease presence*

Problem Statement and Constraints



- *LoRaWAN Bandwidth and Power Constraints*
- *Image transmission over LoRaWAN is not feasible*
- *Battery-powered Sensors & Microcontrollers*

Overall System Architecture



- **Stage1:** *Image Acquisition (VGA 640x480)*
- **Stage2:** *Leaf Localization (Optimized-YOLOv11)*
- **Stage3:** *Disease Classification (MobileNetV2) on cropped leaves*
- **Stage4:** *Result aggregation and LoRaWAN Transmission*

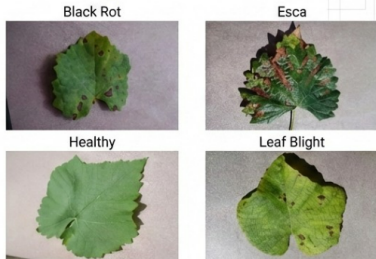
Dataset Preparation and Model Selection

Stage 1: Localization



Localization Dataset: complex vineyard images (Annotated 'Leaf').

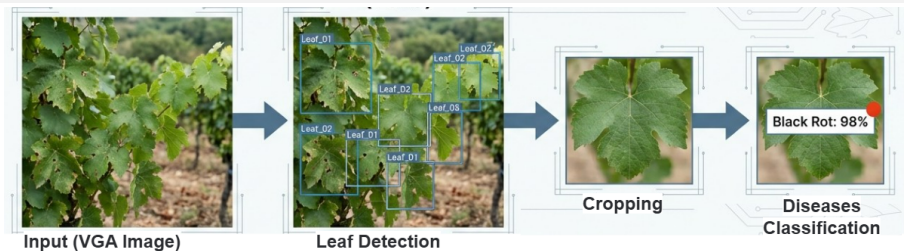
Stage 2: Classification



Classification Dataset: PlantVillage (images, 4 classes).

- **Why Split?** *Separation allows independent optimization on detection and classification tasks*

Inference Flow Design



- **Two-Model Inference Pipeline**
- *Leaf Detection (Optimized-YOLOv11) on full VGA frames*
- *Disease Classification (MobileNetV2) on cropped leaf images*

Conclusion and Future Work

- **Conclusion:** *Successfully demonstrated fully embedded visual leaf disease detection on MCU*
- **Key Contributions:** *Validate the inference pipeline (Capture → Detect → Crop → Classify) on ESP32-S3*
- **Future Work:** *Field testing, model optimization, and expanding to more diseases*

Thank You

Questions?